Normal R.V

$X \sim N(\mu, \sigma^2)$.

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad x \in \mathbb{R}.$$

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1.$$

Standard normal R.V

X is said to be standard normal if

$X \sim N(0, 1)$.

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx = \int_{-\infty}^{\infty} x \frac{e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}}{\sqrt{2\pi}\,\sigma}\,dx$$

$$y = \frac{x-\mu}{\sigma}$$

$$\Rightarrow x = \sigma y + \mu$$

$$dx = \sigma\,dy$$

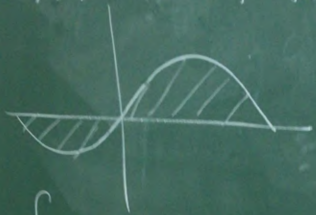
$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\,\cancel{\sigma}} (\sigma y + \mu) e^{-\frac{y^2}{2}} \cancel{\sigma}\,dy$$

$$= \sigma \int_{-\infty}^{\infty} \underbrace{\boxed{\frac{1}{\sqrt{2\pi}}\, y\, e^{-y^2/2}}}_{= f(y)} dy + \mu \boxed{\int_{-\infty}^{\infty} \frac{e^{-y^2/2}}{\sqrt{2\pi}}\, dy}$$

$$= \quad 0 \quad + \mu$$

$f(x)$ is odd if $f(x) = -f(-x)$.

$f(x)$ is even if $f(x) = f(-x)$.



for any odd function $\int_{-\infty}^{\infty} f(y)\, dy = 0$.

$$\mathrm{Var}(X) = E\left[(X-E[X])^2\right]$$

$$= E\left[(X-\mu)^2\right]$$

$$= \int_{-\infty}^{\infty} (x-\mu)^2 f_X(x)\,dx$$

$$= \int_{-\infty}^{\infty} \frac{(x-\mu)^2}{\sqrt{2\pi}\,\sigma} e^{-\left(\frac{x-\mu}{\sigma}\right)^2 \frac{1}{2}} dx \qquad \boxed{\frac{x-\mu}{\sigma} = y}$$

$$= \int_{-\infty}^{\infty} \frac{\sigma^2 y^2}{\sqrt{2\pi}\,\cancel{\sigma}} e^{-\frac{y^2}{2}} \cancel{\sigma}\, dy = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sigma^2 \underbrace{y}_{u(y)}\, \underbrace{y\, e^{-y^2/2}}_{v'(y)}\, dy$$

$$\int_a^b u(x)v'(x)\,dx = u(x)v(x)\Big|_a^b - \int_a^b u'(x)v(x)\,dx \to ①$$

$$\int_a^b u(x)\,v(x)\,dx = \left[u(x)\int v(t)\,dt\right]_a^b - \int_a^b u'(x)\left[\int^x v(t)\,dt\right]dx$$

Suppose $\boxed{v(y) = -e^{-y^2/2}}$

$$v'(y) = y\,e^{-y^2/2}$$

From ①

$$= \frac{1}{\sqrt{2\pi}}\left[\overbrace{\sigma^2 y\,e^{-y^2/2}}^{u(y)v(y)}\Big|_{-\infty}^{\infty} - \underbrace{\int_{-\infty}^{\infty}\sigma^2 \times 1 \times \left(-e^{-y^2/2}\right)dy}_{u'(y)v(y)}\right]$$

$$= 0 + \sigma^2\boxed{\int_{-\infty}^{\infty}\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dy} = r^2$$

$$= \begin{cases} P\left(X \le \frac{y-b}{a}\right) & a > 0 \\ P\left(X \ge \frac{y-b}{a}\right) & a < 0 \end{cases}$$

$$= \begin{cases} F_X\left(\frac{y-b}{a}\right) & \text{if } a > 0 \\ 1 - F_X\left(\frac{y-b}{a}\right) & \text{if } a < 0 \end{cases}$$

$$\frac{\partial F_Y(y)}{\partial y} = \begin{cases} \frac{\partial F_X\left(\frac{y-b}{a}\right)}{\partial y} = \frac{1}{a} f_X\left(\frac{y-b}{a}\right) & a > 0 \\ -\frac{1}{a} f_X\left(\frac{y-b}{a}\right) & a < 0 \end{cases}$$

$$= \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

Lemma: Sc

Proof:

Lemma: Scaling and shifting of any normal R.V results in a normal R.V

$X \sim N(\mu, \sigma^2)$

$Y = aX + b$

$E[Y] = aE[X] + b$
$= a\mu + b$

$Var(Y) = a^2 Var(X)$
$= a^2 \sigma^2$

Then $Y \sim N(a\mu + b, a^2\sigma^2)$

Proof:

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right) \text{ from earlier claim}$$

$$= \frac{1}{|a|} \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2}\left[\frac{\left(\frac{y-b}{a}\right)-\mu}{\sigma}\right]^2}$$

$$\Rightarrow Y \sim N(a\mu + b, a^2\sigma^2)$$

CDF of standard normal R.V

$X \sim N(0,1)$

Let $\Phi(x) = F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \, dt$

Any normal R.V $Y \sim N(\mu, \sigma^2)$ can be thought of as scaling & shifting of standard normal R.V

Suppose $Y = aX + b$ [we want to find a, b]

$E[Y] = aE[X] + b$
$= a \times 0 + b \Rightarrow b = \mu$

$Var(Y) = a^2 Var(X)$
$\Rightarrow \sigma^2 = a^2 \Rightarrow a = \sigma$

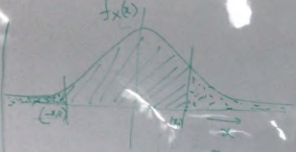
$Y = \sigma X + \mu$

and

$X = \frac{Y - \mu}{\sigma}$

PDF of Standard Normal R.V $X \sim N(0,1)$

$$\phi(x) = 1 - \Phi(-x)$$



$$\Phi(x) = \int_{-\infty}^{x} f(t)\,dt = P(X \le x) \qquad x \geq 0$$

$$= 1 - P(X \geq x) \qquad P(X \geq x) = \int_{x}^{\infty} \frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx$$

$$= 1 - P(X \le -x)$$

$$\boxed{\Phi(x) = 1 - \Phi(-x)} \ \forall x \in \mathbb{R}. \qquad P(X \le -x) = \int_{-\infty}^{-x} \frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx$$

$$\Rightarrow \Phi(0) = 1 - \Phi(0) \Rightarrow \Phi(0) = 1/2$$